

After installation of the libraries using above-mentioned commands, you can proceed with the cloning of TF modules, examples, and files using the following commands:

```
git clone https://github.com/tensorflow/
```

```
tensorflow.git
git clone https://github.com/tensorflow/
models
```

After successfully cloning, go to the directory→research→object_detection→detect.py file in Python IDE (see Fig. 3). Now understand and change the code.

The first part of the code has the required library and modules, as shown in Fig. 4. We have to set the path of the text file that has the category names of species/animals for detection. Here we have set the category labels in New.pbtxt file, as shown in Fig. 5. The



Fig. 9: Detected animals

```
Paragraphs insert Editing
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
'zebra'}}06:02:30PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:31PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:33PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:34PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:36PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:39PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:41PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:43PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:45PM on April
03,2020-----[{'id': 24, 'name':
'zebra'}}06:02:47PM on April
03,2020-----[{'id': 24, 'name':
```

Fig. 10: The contents of forestdb.txt file

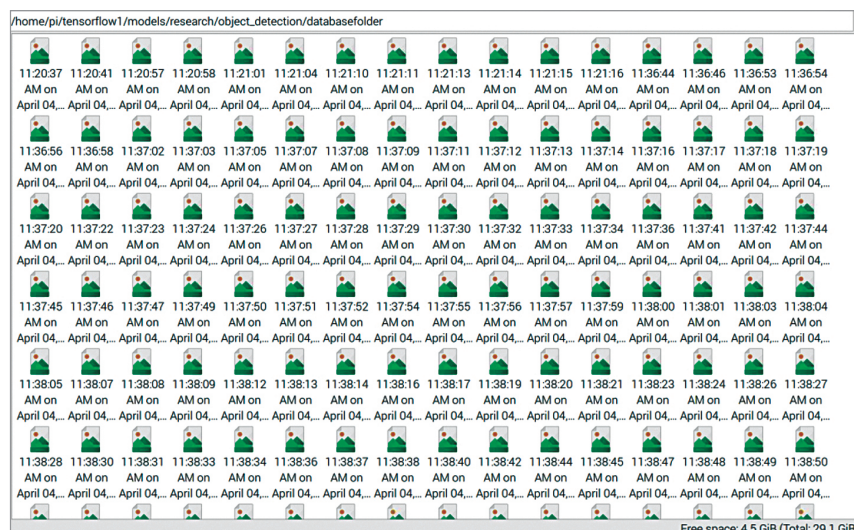


Fig. 11: The images of detected animals stored in databasefolder

next part of the code stores the output of detection result in a string variable called 'NAME.' Then we have the code that marks the detected image name with date and time.

Next, we have a conditional statement that checks the output result of detection. If the condition of the result of detection is not '[']' then it saves the result with date and time in a text file. The next part of the code saves the detected image in 'databasefolder' folder. Here you can change the path for image storage as per your choice (see Fig. 6) and then save the code.

Go to the data folder and create a file named New.pbtxt (see Fig. 7) and label with IDs for different animal species for detection and display (see Fig. 8).

Testing

After finishing the coding part, connect the camera module to Raspberry Pi camera port using a ribbon cable, as shown in the prototype. Open the code in Python IDE and run the script *forestsurveyfinal2.py* and wait for a few minutes so that it can load all the tensor flow modules. This will open a new window with the camera video output.

Bring the camera in front of the animal such as a cow, sheep, etc to be identified. Whenever an animal appears in the camera frame, the detected image will be processed by the computer and saved in a text file. It will save that frame in the databasefolder with date and timestamp. Animals captured by the camera in photo frame are shown in Fig. 9.

Open the *forestdb.txt* file to see the list of species/animals detected along with time and date (see Fig. 10). If you want to see the images/pictures of those detected animals, open databasefolder and click on each image file having the date and time of capture (see Fig. 11).

EFY

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